

Chapter 6

Differential Rules(Part 1)

§ Section: 3.1-3.2

6.1 Derivatives of Basic Functions

The Power Rule

If n is a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

The Power Rule(General Version)

If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example 1.1

Compute $\frac{d}{dx}(x^n)$ where $n \in \mathbb{N}$.

Remark:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \text{ where } \binom{n}{0} = \binom{n}{n} = 1 \text{ and } \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

sol :

$$\begin{aligned} \frac{d}{dx}(x^n) &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{\left(\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k \right) - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{x^n} + n \cdot x^{n-1} h + \binom{n}{2} x^{n-2} h^2 + \dots + h^n - \cancel{x^n}}{h} \\ &= \lim_{h \rightarrow 0} n \cdot x^{n-1} + \binom{n}{2} x^{n-2} h + \dots + h^{n-1} = n \cdot x^{n-1} \end{aligned}$$

Example 1.2

1. $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = (-1)x^{-2} = -\frac{1}{x^2}$
2. $\frac{d}{dx}(\sqrt[3]{x}) = \frac{d}{dx}(x^{\frac{1}{3}}) = \frac{1}{3} \cdot x^{-\frac{2}{3}} = \frac{1}{3} \frac{1}{\sqrt[3]{x^2}}$
3. $\frac{d}{dx}(x^{\sqrt{2}}) = \sqrt{2} \cdot x^{\sqrt{2}-1}$

Example 1.3

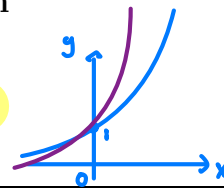
Compute $f'(x)$, where $f(x) = c$.

Sol: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0$

Derivative of the Natural Exponential Function

Given $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$

$\frac{d}{dx}(e^x) = e^x$
 $a > 1, a^x$

**Example 1.4**

Find $f'(x)$ where $f(x) = a^x$.

Sol: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h}$

$$= \lim_{h \rightarrow 0} \frac{a^x a^h - a^x}{h} = \lim_{h \rightarrow 0} a^x \left(\frac{a^h - 1}{h} \right)$$

$$= a^x \cdot \left(\lim_{h \rightarrow 0} \frac{a^h - a^0}{h} \right) = a^x \cdot f'(0) = f(x) \cdot f'(0)$$

Derivative of Exponential Functions

For an exponential function $f(x) = a^x$, $f'(x) = f'(0)a^x = f'(0)f(x)$.

The Constant Multiple Rule

If c is a constant and f is a differentiable function, then

$$[cf(x)]' = cf'(x) \quad \checkmark$$

The Sum and Difference Rule

If f and g are both differentiable, then

$$[f(x) \pm g(x)]' = f'(x) \pm g'(x) \quad \checkmark$$

Example 1.5

Prove that Sum Rule: If $f(x)$ and $g(x)$ are both differentiable, then $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$.

$$\begin{aligned} \text{Sol: } (f(x) + g(x))' &= \lim_{h \rightarrow 0} \frac{(f(x+h) + g(x+h)) - (f(x) + g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{(f(x+h) - f(x)) + (g(x+h) - g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} = f'(x) + g'(x) \end{aligned}$$

Example 1.6

$P(x) = \sum_{k=0}^n a_k x^k$. Find $P'(x)$.

$$\begin{aligned} \text{Sol: } P'(x) &= \frac{d}{dx} \left(\sum_{k=0}^n a_k x^k \right) = \sum_{k=0}^n \frac{d}{dx} (a_k x^k) \\ &= \sum_{k=0}^n a_k \frac{d}{dx} (x^k) = \sum_{k=0}^n a_k k \cdot x^{k-1} = \sum_{k=1}^n a_k k x^{k-1} \end{aligned}$$

Example 1.7

$f(x) = \frac{2}{\sqrt{x}} + 3 - e^{x+1}$. Find $f'(x)$.

$$= 2 \cdot x^{-\frac{1}{2}} + 3 - e \cdot e^x \quad f'(x) = -x^{-\frac{3}{2}} - e \cdot e^x$$

The Product Rule

If f and g are both differentiable, then

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x). \quad \checkmark$$

Example 1.8

Prove that if f and g are both differentiable, then $[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$.

$$\begin{aligned} \text{Sol: } (f(x) \cdot g(x))' &= \lim_{h \rightarrow 0} \frac{(f(x+h)g(x+h)) - (f(x)g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h)g(x+h) - f(x)g(x+h)] + [f(x)g(x+h) - f(x)g(x)]}{h} \\ &= \lim_{h \rightarrow 0} g(x+h) \left(\frac{f(x+h) - f(x)}{h} \right) + f(x) \left(\frac{g(x+h) - g(x)}{h} \right) \\ &= g(x)f'(x) + f(x)g'(x) \end{aligned}$$

Example 1.9

$f(x) = \left(\frac{1}{x} - \frac{1}{x^2}\right)(\sqrt{x^3} + 2)$, Find $f'(x)$.

$$\begin{aligned} f' &= \left(\frac{1}{x} - \frac{1}{x^2}\right)'(\sqrt{x^3} + 2) + \left(\frac{1}{x} - \frac{1}{x^2}\right)(\sqrt{x^3} + 2)' \\ &= \left(-\frac{1}{x^2} + \frac{2}{x^3}\right)(\sqrt{x^3} + 2) + \left(\frac{1}{x} - \frac{1}{x^2}\right) \cdot \frac{3}{2}\sqrt{x} \end{aligned}$$

Example 1.10

$f(x) = e^{2x}$, Find $f'(x)$.

$$= e^x \cdot e^x$$

$$\text{Sol: } f'(x) = e^x \cdot e^x + e^x \cdot e^x = 2 \cdot e^{2x}$$

Example 1.11

$f(x) = \frac{e^{2x}}{\sqrt{x}}$, Find $f'(x)$.

$$= e^{2x} \cdot x^{-\frac{1}{2}}$$

$$f'(x) = 2 \cdot e^{2x} \cdot x^{-\frac{1}{2}} + e^{2x} \cdot \left(-\frac{1}{2}\right) \cdot x^{-\frac{3}{2}} = 2 \frac{e^{2x}}{\sqrt{x}} - \frac{1}{2} \frac{e^{2x}}{\sqrt{x^3}}$$

Example 1.12

$$\begin{aligned}\frac{d}{dx}(f(x)g(x)h(x)) &= (f(x)g(x))' h(x) + (f(x)g(x)) \cdot h'(x) \\ &= (f'(x)g(x) + f(x)g'(x)) \cdot h(x) + f(x)g(x)h'(x) \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)\end{aligned}$$

Example 1.13

$$\begin{aligned}\frac{d}{dx}[(f(x))^2] &= f'(x) \cdot f(x) + f(x) \cdot f'(x) \\ &= 2f(x) \cdot f'(x).\end{aligned}$$

Exercise 1.1

1. For $n \in \mathbb{N}$, compute $\frac{d}{dx}[(f(x))^n] = n(f(x))^{n-1} \cdot f'(x)$
2. For $n \in \mathbb{N}$, compute $\frac{d}{dx}e^{nx}$.

Example 1.14

$f(x) = (x^e - \frac{1}{x^2} + e^x)^{2025}$. Find $f'(x)$.

$$f'(x) = 2025 \left(x^e - \frac{1}{x^2} + e^x \right)^{2024} \cdot \left(e \cdot x^{e-1} + \frac{2}{x^3} + e^x \right)$$

The Quotient Rule

If f and g are both differentiable, then

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

**Example 1.15**

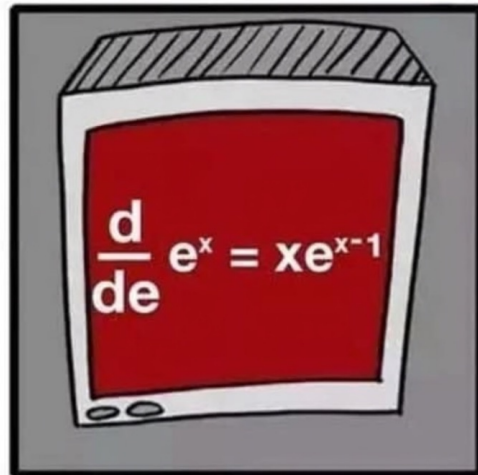
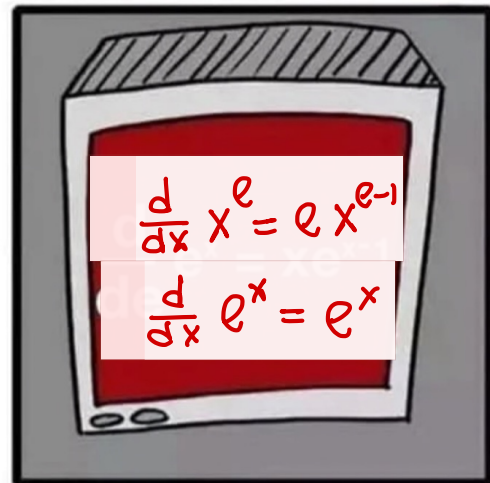
Prove the Quotient Rule.

sol: Let $h(x) = \frac{f(x)}{g(x)}$. Then $h(x) \cdot g(x) = f(x)$.

$$\Rightarrow \frac{d}{dx} h(x) \cdot g(x) + h(x) \cdot g'(x) = f'(x)$$

$$\frac{f'g - fg'}{g^2}$$

$$\Rightarrow h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} = \frac{f'(x) - \frac{f(x)}{g(x)} g'(x)}{g(x)}$$



$$\begin{aligned} \left(\frac{f(x)}{g(x)} \right)' &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{h g(x+h)g(x)} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h)g(x) - f(x)g(x)] + [f(x)g(x) - f(x)g(x+h)]}{h g(x+h)g(x)} \end{aligned}$$

Example 1.16 $\frac{d}{dx} \left(\frac{1}{g(x)} \right) = \frac{-g'(x)}{(g(x))^2} \quad \checkmark$
 Compute $\frac{d}{dx} \left(\frac{1}{g(x)} \right)$ where $g(x)$ is differentiable.

$$= \frac{0 \cdot g(x) - 1 \cdot g'(x)}{(g(x))^2} = \frac{-g'(x)}{(g(x))^2}$$

Exercise 1.2

For $n \in \mathbb{N}$, compute $\frac{d}{dx} \frac{1}{(g(x))^n}$.

Example 1.17

$$f(x) = \frac{3x-1}{x+2}, \quad f'(x) = \frac{3 \cdot (x+2) - (3x-1) \cdot 1}{(x+2)^2}$$

$$= 3 - \frac{7}{x+2}$$

$$\Rightarrow f'(x) = \frac{7}{(x+2)^2}$$

Example 1.18

$$f(x) = \frac{x^4 - 3x^3 + 5x}{x^6 - 2x^3 + 6x - 4}. \quad \text{Compute } f'(0). = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{x^3 - 3x^2 + 5}{x^6 - 2x^3 + 6x - 4} = -\frac{5}{4}.$$